BTS–III(R/S)–11–23–2153 Reg. No. 2 0 6 2 2 0 2 7

C

B.Tech. Degree III Semester Regular/Supplementary Examination November 2023

SE 19-206-0303 ENGINEERING FLUID MECHANICS AND MACHINERY

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Analyse fluid flow behaviour under different operating conditions.
CO2: Apply equations of motion to flow problems.
CO3: Estimate the major and minor losses associated with fluid flow in piping networks.
CO4: Select hydraulic machines based on the type of operation.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	Define capillarity. The capillary rise in a glass tube is not to exceed 0.2 mm of water. Determine its minimum size, given that surface tension of water in contact with air is 0.0725 N/m.	3	L2	1	1,2
	(b)	What are the types of fluid flows? Define each of them.	3	L1	1	1
	(c)	Define velocity potential function and stream function.	3	L1	2	1
	(d)	State the conditions of equilibrium of sub-merged bodies.	3	L2	2	1
	(e)	Classify boundary layer in fluid flow.	3	L1	3	1
	(f)	What does Hagen- Poiseuille equation refer to?	3	L2	3	1
	(g)	Define cavitation in pump. What is the need for priming in pumps?	3	L2	4	1
	(h)	What is speed ratio and specific speed in turbines?	3	L1	4	1

PART B

		(4 × 12 = 48)				
II.	(a)	Derive expression for total pressure and centre of pressure for a vertically immersed surface.	6	L2	1	
	(b)	Explain the types of instruments used to measure pressure.	6	L1	1	
		OR				
III.		A U-tube manometer is used to measure the pressure in a pipeline, which is in excess of atmospheric pressure. The right limb of the manometer is open to atmosphere. The contact between mercury and water is in the left limb. Determine the pressure of water in the pipeline, if the difference in level of mercury in the manometer is 10 cm and the free surface of mercury is in level with the centre of pipe. Sketch the arrangement.	12	L3	1	1,2

(P.T.O.)

			Marks	BL	CO	PO
IV.	(a)	The following cases represent the two velocity components. Determine the third component of velocity such that they satisfy the continuity equation: (i) $u = x^2 + y^2 + z^2, v = xy^2 - yz^2 + xy$ (ii) $v = 2y^2, w = 2xyz$.	6	L2	2	1,2
	(b)	What is total acceleration and velocity in three-dimensional flow?	6	L1	2	1
		OR				
V.	(a)	Derive Euler's equation of motion along a streamline for an ideal fluid and obtain Bernoulli's equation from it. State all the assumptions made.	6	L2	2	1
	(b)	What is a venturi meter? Derive an expression for the rate of flow of fluid through it.	6	L1	2	1
VI.	(a)	What are the minor losses in pipe flow?	4	L1	3	1
	(b)	An oil of viscosity 0.1 Ns/m^2 and relative density 0.9 is flowing through a circular pipe of diameter 50 mm and of length 300 m. The rate of flow of fluid through the pipe is 3.5 litre/s. Find the pressure drop in a length of 300 m and also the shear stress at the pipe wall.	8	L3	3	1,2
		OR				
VII.		An oil of specific gravity 0.9 and viscosity 0.06 poise is flowing through a pipe of diameter 200 mm at the rate of 60 litre / sec. Find the head loss due to friction for a 500 m length of pipe. Find the power required to maintain this flow.	12	L3	3	1,2
VIII.		Explain the major parts and their functions of a centrifugal pump with a neat sketch.	12	L1	4	1
		OR				
IX.		With the help of neat diagram explain the construction and working of a radial flow reaction turbine.	12	L1	4	1

Blooms's Taxonomy Levels
L1 – 48.33%, L2 – 25%, L3 – 26.66%.

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